THE DIAGONAL OF THE STACK OF POLARIZED SCHEMES IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file quot.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Lemma quot-lemma-polarized-to-spaces-algebraic (canonical tag 0D41). cited at proof line 4278. Collected card: representable morphism to the stack of spaces.
- Lemma quot-lemma-spaces-diagonal (canonical tag 0D1F). cited at proof line 4279. Collected card: diagonal of the stack of spaces.
- Lemma criteria-lemma-diagonals-and-algebraic-morphisms (canonical tag 0D3R). cited at proof line 4281. Diagonals criterion.

Source reference. Original lemma 0D41: sources/stacks-chow/tags.txt, source lines 16436–16436. Cited at statement line 4278: representable morphism to the stack of spaces.

Source reference. Original lemma 0D1F: sources/stacks-chow/tags.txt, source lines 16345–16345. Cited at statement line 4279: diagonal of the stack of spaces is representable.

Source reference. Original lemma 0D3R: sources/stacks-chow/tags.txt, source lines 16426–16426. Cited at statement line 4281: diagonals-and-algebraic criterion.

Lemma 0.2. *The diagonal*

$$\Delta : \mathcal{P}olarized \longrightarrow \mathcal{P}olarized \times \mathcal{P}olarized$$

is representable by algebraic spaces.

Proof. This is a formal consequence of Lemmas 0D41 and 0D1F. See Criteria for Representability, Lemma 0D3R. □

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.